

Experiment	<i>Lab 02: Experiment AFL – Lift and Drag on Airfoils</i>
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Summary

This experiment's main objective is to evaluate an airfoil's aerodynamic performance. The experiment involved measuring the distribution of surface pressure around the airfoil and using a pitot static tube to do a wake survey in order to calculate the drag on the airfoil. Making use of the data, the experimental performance parameters were computed to determine the airfoil's aerodynamic performance. These features included the moment coefficients and the coefficients of lift and drag. Potential flow predictions from a theoretical solution were contrasted with the computed findings. Plotting of the attributes revealed variations between the experimental and theoretical values. Numerous factors could have contributed to the discrepancies between the experimental and theoretical values. Viscosity effects are one example of this variable. The panel code's performance values are significantly higher since it operates in a controlled environment free from external influences. Experimental lift values were less than possible flow predictions due to viscous factors.

Aim & Objectives

The aim of this laboratory is to measure an airfoil's wake profile and surface pressure distribution in order to assess its aerodynamic performance and ascertain its lift and drag characteristics.

The objectives of the lab are:

1. Record the distributions of surface pressure surrounding the airfoil.
2. Using the traversing pitot static tube, perform a wake survey.
3. Determine the coefficients of lift, drag, and moment.
4. Analyze the experimental findings in relation to potential flow.
5. Determine the zero lift angle of attack.

Uncertainty Analysis

The uncertainty of the velocity had to be determined before the uncertainty of other wing parameters propagated. This formula is used to determine the velocity's uncertainty:

$$\delta V = \delta V_{p_{dyn}} + \delta V_{p_{atm}} + \delta V_{T_{atm}}$$

In this case, δV represents the velocity uncertainty, $\delta V_{p_{dyn}}$ represents the dynamic pressure uncertainty, $\delta V_{p_{atm}}$ represents the atmospheric pressure uncertainty, and $\delta V_{T_{atm}}$ represents the atmospheric temperature uncertainty. Each of the following formulas is used to determine the other uncertainty values:

$$\delta V_{P_{dyn}} = \sqrt{\frac{2 * (P_{dyn} + \delta P_{dyn})}{\frac{P_{atm}}{R * (T_{atm})}}} - V$$

$$\delta V_{P_{atm}} = \sqrt{\frac{2 * P_{dyn}}{\frac{(P_{atm} + \delta P_{atm})}{R * (T_{atm})}}} - V$$

$$\delta V_{T_{atm}} = \sqrt{\frac{2 * P_{dyn}}{\frac{P_{atm}}{R * (\delta T_{atm})}}} - V$$

Now, we calculate the uncertainty of C_p :

$$\delta C_p = \sqrt{\left(\frac{\partial C_p}{\partial P_{dyn}} \delta P_{dyn}\right)^2 + \left(\frac{\partial C_p}{\partial P_{atm}} \delta P_{atm}\right)^2 + \left(\frac{\partial C_p}{\partial T_{atm}} \delta T_{atm}\right)^2 + \left(\frac{\partial C_p}{\partial V} \delta V\right)^2}$$

Uncertainty of the coefficient of drag:

$$C_d = \frac{2}{U_\infty^2 C} \int_{-\infty}^{\infty} U_w (U_\infty - U_w) dy$$

$$\frac{\partial C_d}{\partial U_w} = \frac{2}{U_\infty C} \left(1 - \frac{2U_w}{U_\infty}\right)$$

Uncertainty of the coefficient of lift:

$$C_l = \frac{1}{c} \int_{LE}^{TE} (C_{p_l} - C_{p_u}) dx$$

$$\frac{\partial C_l}{\partial C_p} = \frac{1}{c}$$

$$\partial C_l = \frac{\delta C_p}{c}$$

$$\partial C_l = \frac{(\delta C_{p_l} + \partial C_{p_{lmax}}) - (\delta C_{p_u} + \partial C_{p_{umax}})}{c}$$

Uncertainty of the coefficient of moment for the leading edge and quarter chord:

$$C_{MLE} = -\frac{1}{c^2} \int_{LE}^{TE} (C_{p_l} - C_{p_u}) dx$$

$$C_{MC/LE} = -\frac{1}{c^2} \int_{LE}^{TE} (C_{p_l} - C_{p_u}) (x - 0.25) dx$$

Results & Discussion

The goal of this lab was to comprehend the flow around an airfoil at various angles of attack while maintaining a constant speed. Next, contrast the prospective flow around an airfoil with the actual flow.

As can be seen below, the NACA 4412 airfoil in question had 18 static pressure taps spread throughout it.

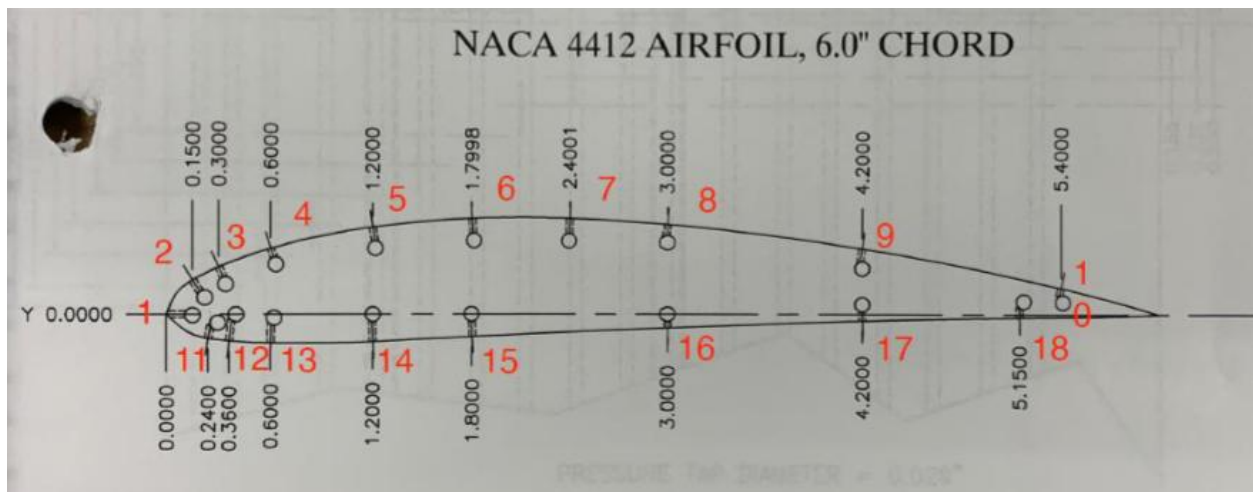


Figure 1: NACA 4412 Airfoil

This gauges the airfoil's pressure distribution. The C_p , C_l , C_d , and C_m for the leading edge and quarter chord, among other things, will then be ascertained by interpolating that data.

It is necessary to gather the ambient pressure and temperature prior to conducting experiments. It was measured that the temperature was 22 C° and the pressure was 30.1 inHg. This is necessary to determine the air's density. The density of air was calculated in Appendix A. We can determine the pressure required in the wind tunnel to reach the desired speed using the air density. 15 m/s was our choice. The following formula is used to calculate the pressure.

$$P_{diff} = \frac{V^2 * \rho_{air}}{2}$$

The code was run after the tunnel was adjusted to this speed, the transverse pitot tube was adjusted to its lowest height, and the airfoil was adjusted to attack at a -3 degree angle. The transverse pitot tube rose in tiny steps to measure the wake distribution as soon as the code began. Additionally, the algorithm used the 18 static ports to measure the pressure surrounding the airfoil. Following data collection, this was done again in steps of three for angles 0 through 12.

Wake Profile

The pressure had to be translated from inches of water to Pascals. It was necessary to translate the height from inches to meters.

Below is the formula to get the wake velocity.

$$V_{wake} = \sqrt{\frac{2 * P_{static}}{\rho_{air}}}$$

To view the wake profile, we will plot the velocity against height for each angle of attack after we know the velocity at each incremental height.

Wake Velocity Profiles

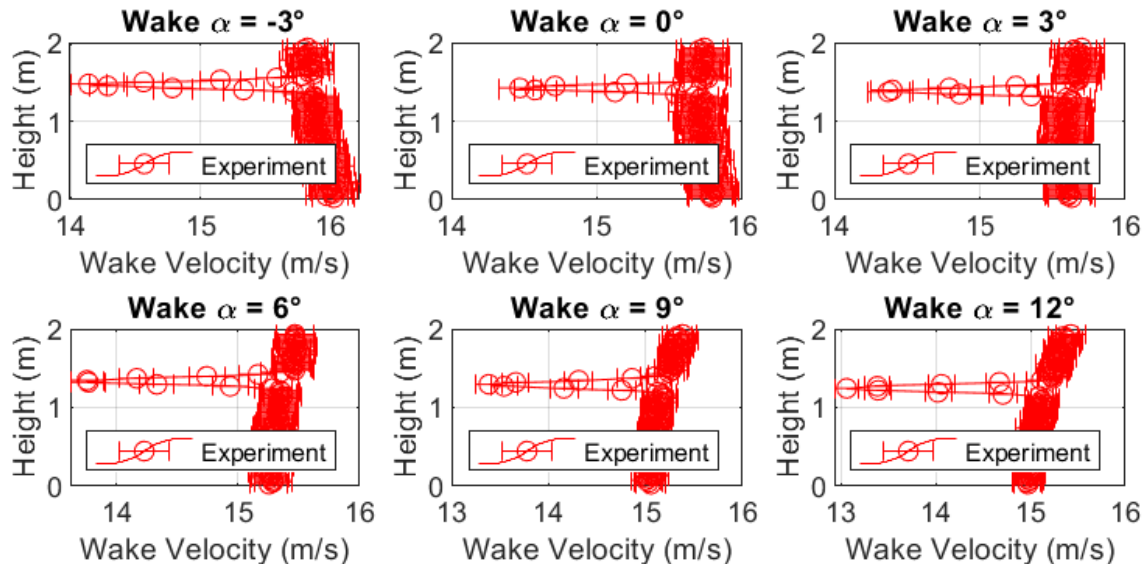


Figure 2: Wake Profile for all AoA

The wake, or turbulent air that emerges from the airfoil, slows down the air when the pitot tube passes behind it, as can be seen. This happens as a result of the flow diverging from the pressure distribution at the airfoil's end, creating counter-rotating vortices and slowing down the air behind it.

The data shows that the wake grows with increasing angles of attack, which leads to the boundary layers separating earlier, increasing drag, and decreasing lift efficiency.

Potential Flow

Let's talk about prospective flow before we examine the remaining data. The data will be compared to the prospective flow, which was estimated using certain hypotheses. These presumptions include that it is inviscid (has no viscosity), incompressible (the fluid density stays constant), and irrotational.

(The fluid doesn't rotate) and constant (The flow characteristics don't alter over time).

Equations with the potential flow assumptions were used to compute the potential flow using the Panel Code. Together with all the data calculations, this can be found in the Appendix and Panel code references.

Cp Distribution

We must first convert from inches of water to Pascals in order to interpret the data gathered from the airfoil's 18 ports. The dynamic pressure must then be determined using the formula below.

$$q = \frac{1}{2} * \rho_{air} * V^2$$

Formula below to calculate the coefficient of pressure:

$$C_p = \frac{P_{diff} + q_{inf}}{q_{inf}}$$

Once the coefficient of pressure for each of the 18 ports is known, we will divide them into the airfoil's top and lower surfaces, plot them against the chord length for each angle of attack, and compare the results to the possible flow.

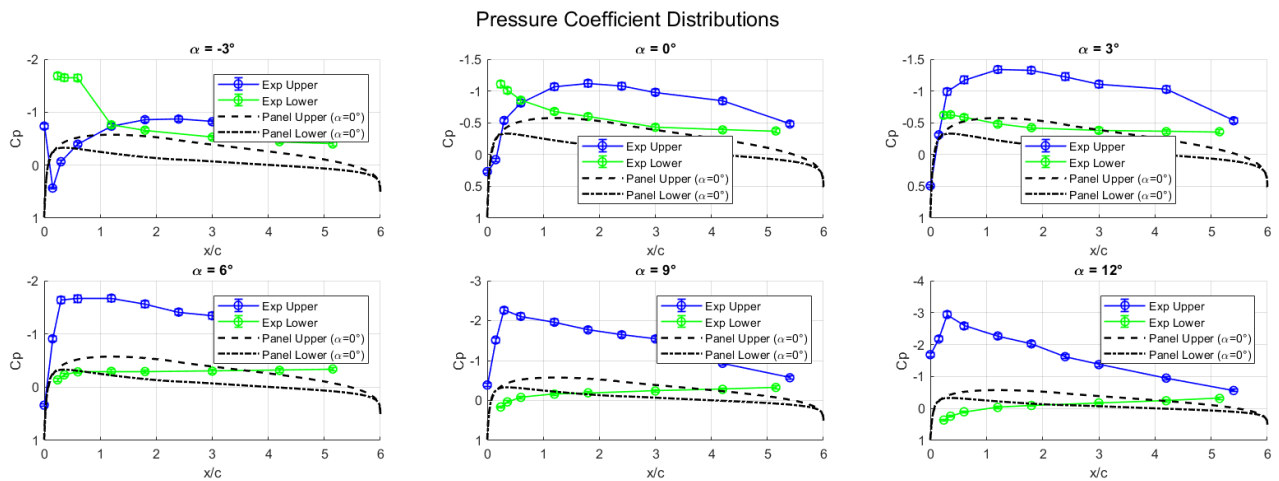


Figure 3: Cp distribution for all Angles of Attack

The primary distinction is that a variety of factors, including skin friction and air rotation, can alter the pressure distribution.

Cl Analysis

We will use the following formula to determine the coefficient of lift for each angle after determining the Cp distribution for each angle of attack.

$$C_l = \frac{1}{c} \int_{LE}^{TE} (C_{p_l} - C_{p_u}) dx$$

We will plot each Cl against the potential flow for comparison.

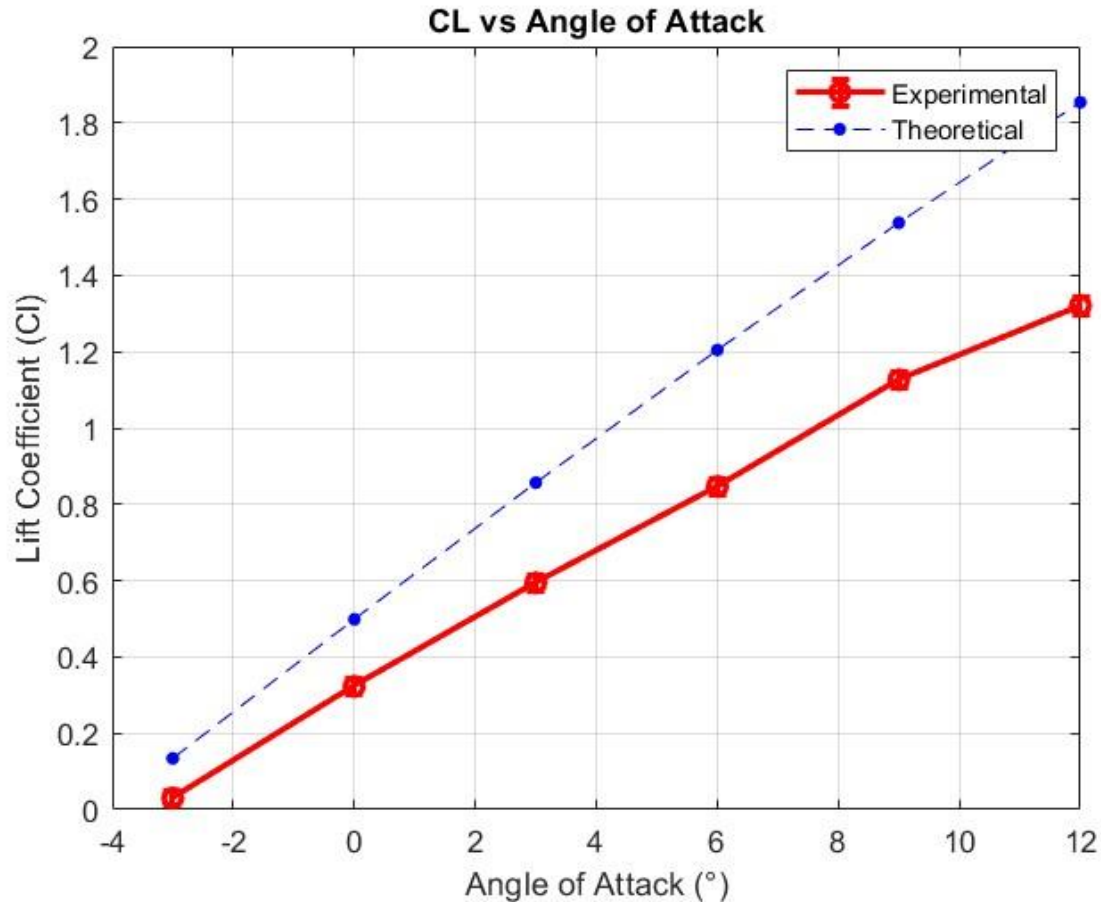


Figure 4: C_l vs Angle of Attack curve for NACA 4412

As can be seen above, we are below the potential flow line due to the influences of the real world, even though we are extremely close to it. All of the presumptions regarding potential flow are no longer valid when skin friction is taken into consideration. Because the pressure differential will be greater at higher angles of attack, we may also observe that as the angle of attack increases, so does the lift.

Cd Analysis

We use the following formula to calculate the coefficient of drag:

$$D = \rho_{air} \int_1^2 U_y (U_{mean} - U_y) dy$$

$U_y \equiv$ mean velocity of the free stream

$U_{mean} \equiv$ velocity of bump in wake profile

$$C_d = \frac{D}{\frac{1}{2} * \rho_{air} * U_{mean}^2 * c}$$

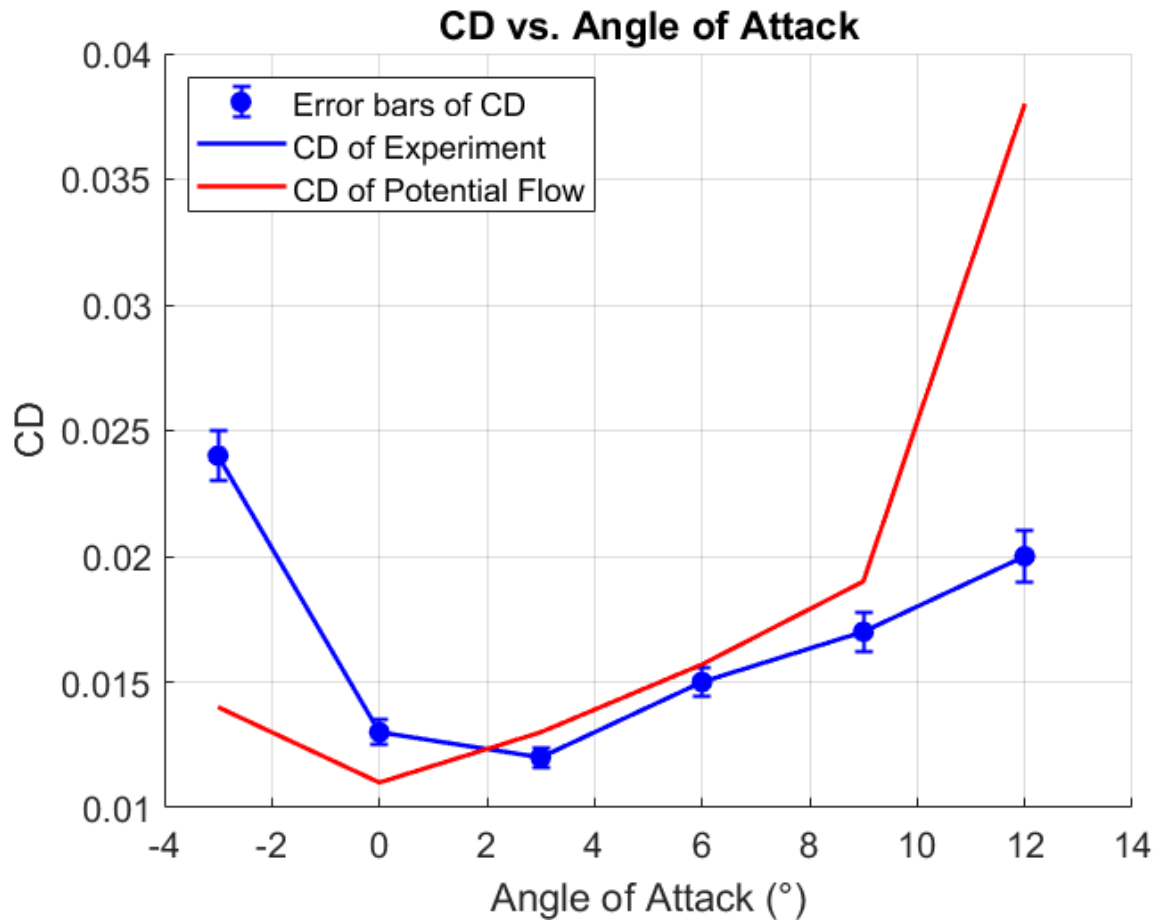


Figure 5: Cd vs Angle of Attack curve for NACA 4412

At the -3 angle, the drag is quite significant, however, this is most likely the result of human error and a larger angle than -3. Possible improvements include the need to take skin friction into consideration and the elimination of all conceivable flow assumptions. Because of the magnitude of the wake increase, the drag rises with the angle of attack. This happens as a result of flow separation on the airfoil previously.

Cm – Leading Edge

We will use the formula below to determine the coefficient of momentum about the leading edge for each angle after determining the Cp distribution for each angle of attack.

$$C_{MLE} = -\frac{1}{c^2} \int_{LE}^{TE} (C_{p_l} - C_{p_u}) dx$$

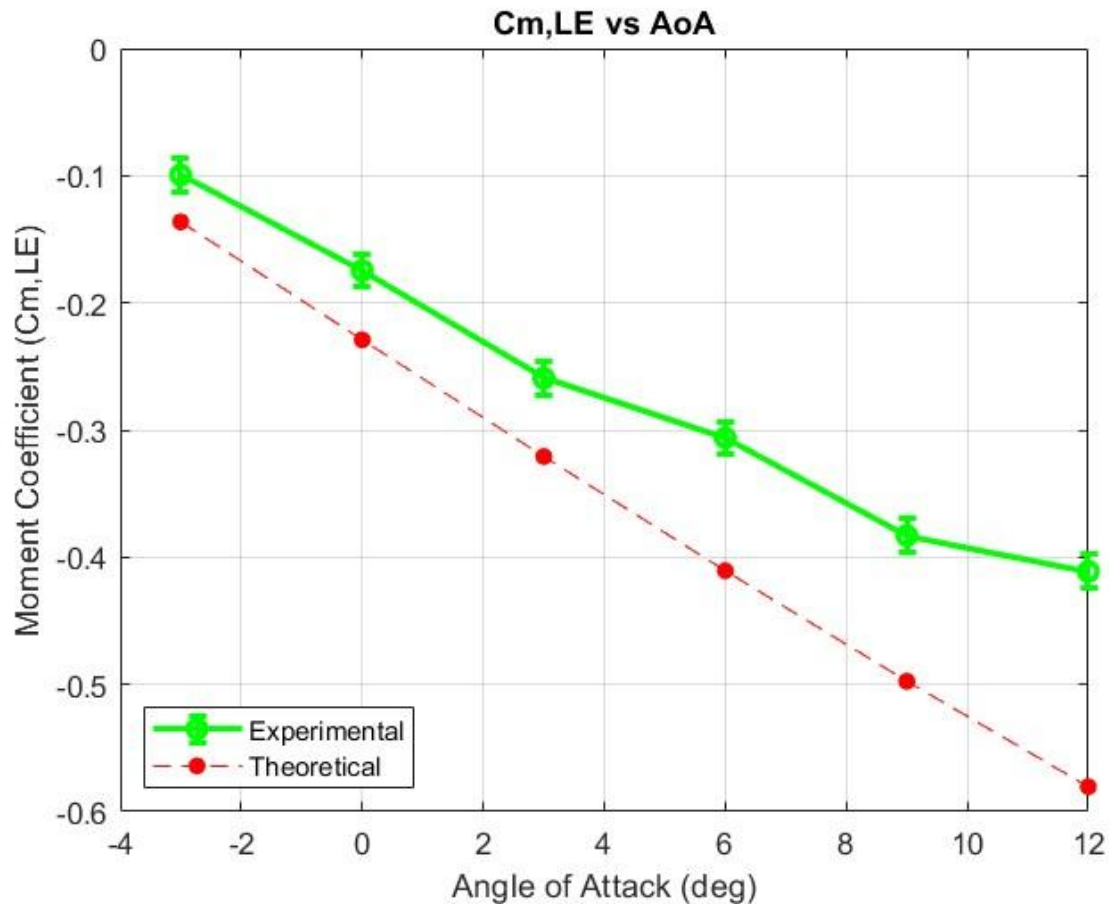


Figure 6: $C_{m,LE}$ vs Angle of Attack curve for NACA 4412

As can be seen above, we are above the potential flow line. All of the presumptions regarding potential flow are no longer valid when skin friction is taken into consideration.

We see a growing negative coefficient of moment about the leading edge as the angle of attack rises. This indicates that as the angle increases, the airfoil's pitching moment decreases.

C_m – Quarter Chord

We will use the formula below to determine the coefficient of momentum about the quarter chord for each angle.

$$C_{MC/LE} = -\frac{1}{c^2} \int_{LE}^{TE} (C_{p_l} - C_{p_u})(x - 0.25) dx$$

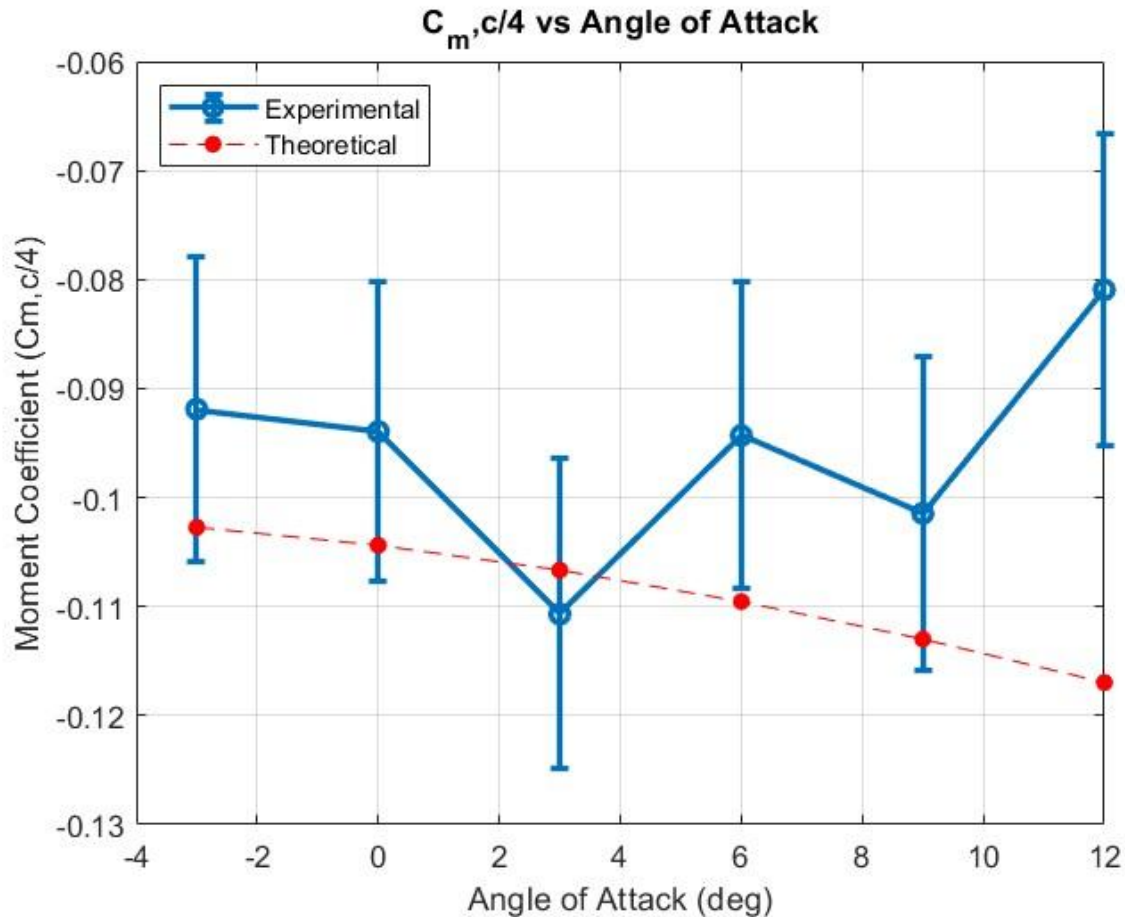


Figure 7: $C_{m,c/4}$ vs Angle of Attack curve for NACA 4412

We are really near the possible flow line, as can be seen above, although it varies with each angle of attack. All of the presumptions for potential flow are no longer valid when skin friction is taken into consideration.

The negative coefficient of moment about the leading edge decreases as the angle of attack rises. This indicates that the airfoil has a negative pitching moment, which diminishes with increasing angle.

Cd vs Cl

Now, we plot the C_d vs C_l curve, also known as the drag polar, and analyze it based on its shape.

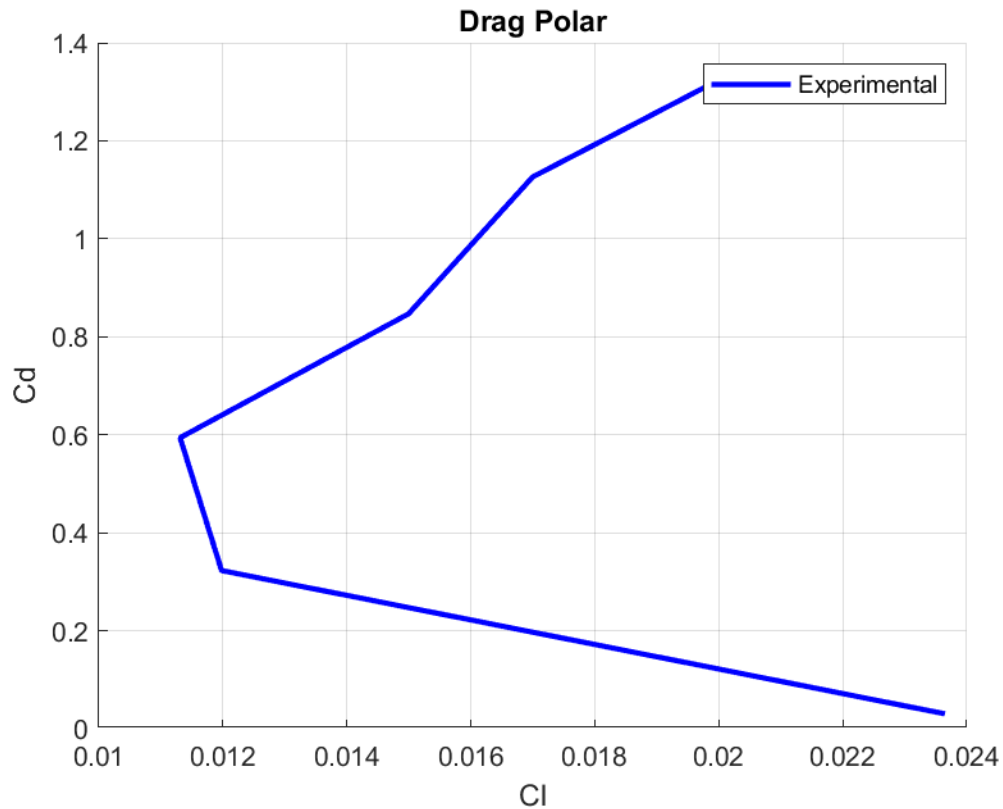


Figure 8: C_l vs C_d for NACA 4412

The relationship between the drag and lift coefficients is displayed by the drag polar plot. This creates what's called the "drag bucket." The experiment is shaped more like a bowl than a fishhook. This, in my opinion, occurs as a result of taking into consideration actual possibilities and the fact that the -3 drag was so strong that there may have been some inaccuracy.

We can see that the airfoil has good aerodynamic drag, but either human error or an angle of attack of -3 caused the airfoil to endure a lot of drag, which lowers the efficiency of life at that angle.

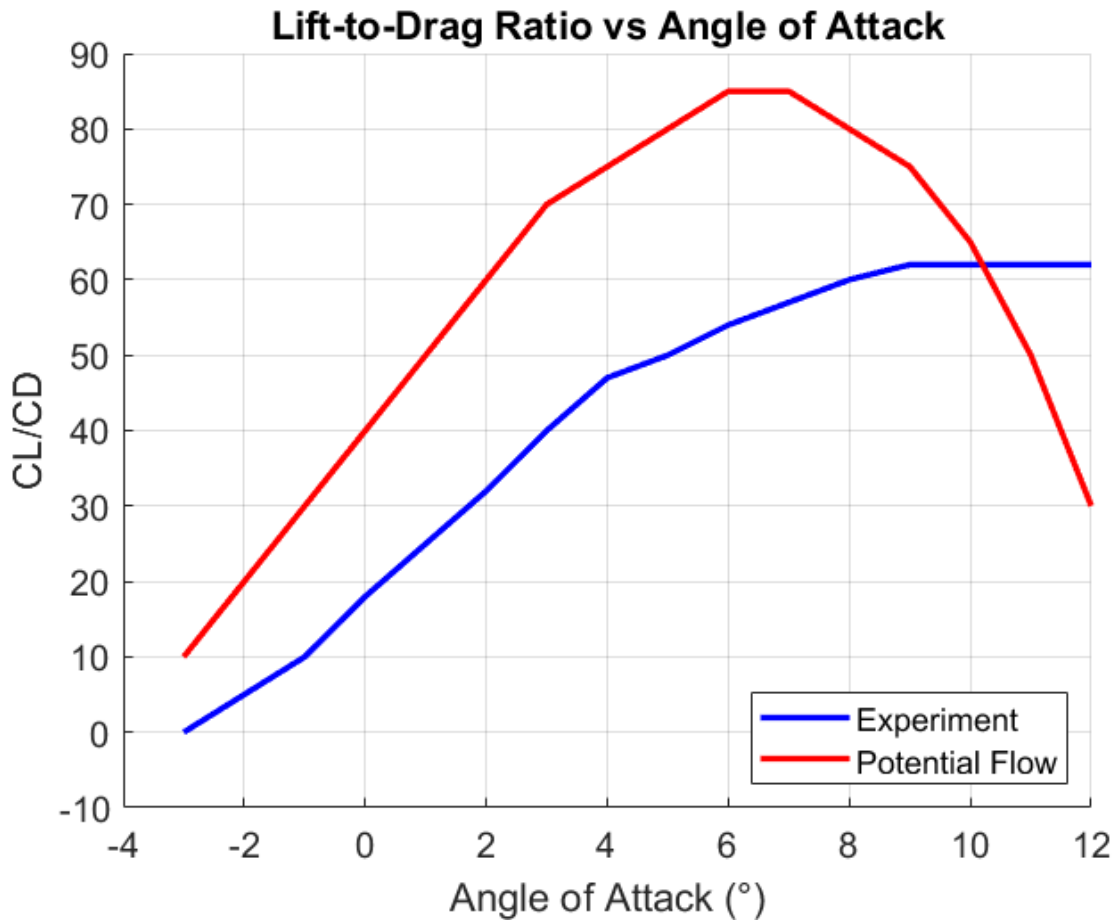


Figure 9: CL/Cd vs Angle of Attack

The airfoil's lift-to-drag ratio rises to a particular point before falling again. This is because not enough lift is generated in response to the increased drag. The graph indicates that at almost 9 degrees AoA, the lift-to-drag ratio is at its maximum.

Zero-Lift Angle of Attack

The final requirement of this lab is to determine the zero-lift angle of attack. Through theoretical equations and using Figure 4, the angle of attack for zero lift was found to be -3.31 degrees.

References

1. Experiment AFL – Lift and Drag on Airfoils: (2025). Embry-Riddle Aeronautical University.
2. AE315 Supplement – Experimental Aerodynamics Supplement: (2025). Embry-Riddle Aeronautical University.

Appendix

Appendix A: Density Calculation

$$P = \rho RT$$

$$\rho_{air} = \frac{P}{RT}$$

$$\rho_{air} = \frac{101930.3}{287 * 295.15} = 1.203 \frac{kg}{m^3}$$

Appendix C: MATLAB Code

```
% Author: Anuranan Bharadwaj
% Lab 03: Experiment AFL
```

```
clc;
clear;
close all;
```

```
% Constants
```

```
R = 287;
Po = 30.2 * 3386.39;      % Pa
To = 295;                 % K
rho0 = Po / (R * To);     % kg/m^3
```

```
%% Traverse Data
```

```
dp = zeros(7, 76);
z_matrix = zeros(7, 76);
AoA_vals = [-3, 0, 3, 6, 9, 12];
```

```
for i = 1:length(AoA_vals)
    k = AoA_vals(i);
    for j = 1:76

        if k < 0
            fileName = sprintf('AoA_n%d_%d.mat', abs(k), j);
        else
            fileName = sprintf('AoA_%d_%d.mat', k, j);
        end

        filePath = fullfile('LAB 3', fileName);

        TraverseData = load(filePath);

        dp(i, j) = mean(TraverseData.dp) * 248.8; % Pascals

        Patm = TraverseData.pAtm * 3386.39;
        Tatm = TraverseData.tAtm;

        z_matrix(i, j) = TraverseData.zCurr * 0.0254; % Meters
    end
end
```

```
%% Pressure Data
```

```
p_mean = zeros(7, 18);
for i = 1:length(AoA_vals)
    AoA = AoA_vals(i);

    if AoA < 0
```

```

        fileName = sprintf('AoA_n3.csv'); testing
        fileName = sprintf('AoA_n%d.csv', abs(AoA));
    else
        fileName = sprintf('AoA_%d.csv', (AoA));
    end

    p_data = readtable(fileName);

    p_mean(i, :) = mean(table2array(p_data(:, 3:20))) * 248.8; % Mean values
end

%% Analysis
Velocity = sqrt(2 * dp / rho0);
PosiVarble = dp + Po;
P_inf = Po - rho0;

UppBnd = [69, 65, 60, 60, 59, 56, 56];
LowBnd = [31, 51, 47, 47, 44, 42, 40];

freestream = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    freestream(i) = mean([Velocity(i, 1:LowBnd(i)), Velocity(i, UppBnd(i):end)]);
end

D = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    D(i) = rho0 * trapz(z_matrix(i, LowBnd(i):UppBnd(i)), ...
        Velocity(i, LowBnd(i):UppBnd(i)) .* (freestream(i) - Velocity(i, LowBnd(i):UppBnd(i))));
end

%% Drag Coefficient
c_length = 6 * 0.0254; % Cord length in meters
Cd = D ./ (0.5 * rho0 * freestream.^2 * c_length);

q = 0.5 * rho0 * freestream.^2;

Cp = zeros(7, size(p_mean, 2));
for i = 1:length(AoA_vals)
    Cp(i, :) = (p_mean(i, :) ./ q(i)) + 1;
end

%% Surface Data
x_pos = [0, 0.15, 0.3, 0.6, 1.2, 1.7998, 2.4001, 3, 4.2, 5.4, ...
    0.24, 0.36, 0.6, 1.2, 1.8, 3, 4.2, 5.15] ./ 6;

upper_x_val = x_pos(1:10);
lower_x_val = x_pos(11:end);
Cp_upper = Cp(:, 1:10);

```

```
Cp_lower = Cp(:, 11:end);

%% Lift and Moment Coefficients
Cl = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    Cl(i) = trapz(lower_x_val, Cp_lower(i, :)) - trapz(upper_x_val, Cp_upper(i, :));
end

Cm_le = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    Cm_le(i) = trapz(upper_x_val, Cp_upper(i, :) .* upper_x_val) - ...
        trapz(lower_x_val, Cp_lower(i, :) .* lower_x_val);
end

Cm_cby4 = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    Cm_cby4(i) = Cl(i) / 4 + Cm_le(i);
end

% Zero-Lift AoA
zero_lift_AoA = interp1(Cl, AoA_vals, 0, 'linear', 'extrap');
fprintf('Zero-Lift Angle of Attack: %.2f°\n', zero_lift_AoA);

% Panel Code Data
panelCode = load('PanelCode.mat');

%% Uncertainty
dP = 16.93;
dT = 0.5;
dz = 0.005 * 0.0254;
dV = 0.01;

% Density
rho = rho0 * sqrt((dP/Po)^2 + (dT/To)^2);

% Dynamic pressure
dq = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    dq_rho = 0.5 * freestream(i)^2;
    dq_dV = rho0 * freestream(i);
    dVel = dV * freestream(i);
    dq(i) = sqrt((dq_rho * rho)^2 + (dq_dV * dVel)^2);
end

% Cd
dCd = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    dVwake = dV * Velocity(i, LowBnd(i):UppBnd(i));
    dD_rho = trapz(z_matrix(i, LowBnd(i):UppBnd(i)), Velocity(i, LowBnd(i):UppBnd(i)) ...
        (i)) .* ...
```

```

        (freestream(i) - Velocity(i, LowBnd(i):UppBnd(i))));
    dD_dV = rho0 * trapz(z_matrix(i, LowBnd(i):UppBnd(i)), (freestream(i) - 2 * Velocity(i, LowBnd(i):UppBnd(i))));
    dD = sqrt((dD_drho * drho)^2 + (dD_dV * mean(dVwake))^2);

    dCd_dD = 1 / (0.5 * rho0 * freestream(i)^2 * c_length);
    dCd_dq = -D(i) / (0.5 * rho0 * freestream(i)^2 * c_length)^2;
    dCd(i) = sqrt((dCd_dD * dD)^2 + (dCd_dq * dq(i))^2);
end

% Cp
dCp = zeros(7, size(p_mean, 2));
for i = 1:length(AoA_vals)
    dCp_dp = 1 / q(i);
    dCp_dq = -p_mean(i, :) / q(i)^2;
    dPresScn = 0.01 * abs(p_mean(i, :)) + 248.8 * 0.005;
    dCp(i, :) = sqrt((dCp_dp * dPresScn).^2 + (dCp_dq * dq(i)).^2);
end
dCp_upper = dCp(:, 1:10);
dCp_lower = dCp(:, 11:end);

% Cl
dCl = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    dCp_upper_weighted = sqrt(sum((dCp_upper(i, :) .* diff([0, upper_x_val])).^2));
    dCp_lower_weighted = sqrt(sum((dCp_lower(i, :) .* diff([0, lower_x_val])).^2));
    dCl(i) = sqrt(dCp_upper_weighted^2 + dCp_lower_weighted^2);
end

% Cm_Le
dCm_le = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    dCm_upper = sqrt(sum((dCp_upper(i, :) .* upper_x_val .* diff([0, upper_x_val])).^2));
    dCm_lower = sqrt(sum((dCp_lower(i, :) .* lower_x_val .* diff([0, lower_x_val])).^2));
    dCm_le(i) = sqrt(dCm_upper^2 + dCm_lower^2);
end

% Cm_c/4
dCm_cby4 = zeros(1, length(AoA_vals));
for i = 1:length(AoA_vals)
    dCm_cby4_dCl = 1 / 4;
    dCm_cby4_dCm = 1;
    dCm_cby4(i) = sqrt((dCm_cby4_dCl * dCl(i))^2 + (dCm_cby4_dCm * dCm_le(i))^2);
end

% L/D
dLbyD = zeros(1, length(AoA_vals));

```



```

for i = 1:length(AoA_vals)
    LbyD = Cl(i) / Cd(i);
    dLbyD(i) = LbyD * sqrt((dCl(i) / Cl(i))^2 + (dCd(i) / Cd(i))^2);
end

%% Plotting Section
figure;
for i = 1:length(AoA_vals)
    subplot(3, 3, i);
    dVel = dV * Velocity(i, :);
    errorbar(Velocity(i, :), (1:76) * 0.0254, [], [], dVel, dVel, '-ro');
    title(sprintf('Wake \\alpha = %d°', AoA_vals(i)));
    xlabel('Wake Velocity (m/s)');
    ylabel('Height (m)');
    legend('Experiment', 'Location', 'southwest');
    grid on;
end
sgtitle('Wake Velocity Profiles');
saveas(gcf, 'ee_1.png');

% Pressure Distributions
figure;
for i = 1:length(AoA_vals)
    subplot(3, 3, i);
    hold on;
    errorbar(upper_x_val * 6, Cp_upper(i, :), dCp_upper(i, :), '-bo', 'LineWidth', 1, 'DisplayName', 'Exp Upper');
    errorbar(lower_x_val * 6, Cp_lower(i, :), dCp_lower(i, :), '-go', 'LineWidth', 1, 'DisplayName', 'Exp Lower');

    plot(panelCode.xu * 6, panelCode.Cpu, '--k', 'LineWidth', 1.5, 'DisplayName', '
    sprintf('Panel Upper (\\alpha=%d°)', panelCode.alpha));
    plot(panelCode.xl * 6, panelCode.Cpl, '-.k', 'LineWidth', 1.5, 'DisplayName', '
    sprintf('Panel Lower (\\alpha=%d°)', panelCode.alpha));

    set(gca, 'YDir', 'reverse');
    xlabel('x/c');
    ylabel('Cp');
    title(sprintf('\\alpha = %d°', AoA_vals(i)));
    legend('Location', 'Best');
    grid on;
    hold off;
end
sgtitle('Pressure Coefficient Distributions');
saveas(gcf, 'ee_2.png');

% Cl_alpha
theory_Cl = [0.1343, 0.4987, 0.8568, 1.2045, 1.5381, 1.8540];
figure;

```

```

errorbar(AoA_vals, Cl, dCl, '-ro', 'LineWidth', 2, 'DisplayName', 'Experimental');
hold on;
plot(AoA_vals, theory_Cl, '--b.', 'MarkerSize', 15, 'DisplayName', 'Theoretical');
xlabel('Angle of Attack (°)');
ylabel('Lift Coefficient (Cl)');
title('CL vs Angle of Attack');
legend('Location', 'northeast');
grid on;

```

```

% Cd_alpha
figure;
errorbar(AoA_vals, Cd, dCd, '-ro', 'LineWidth', 4, 'DisplayName', 'Experimental');
xlabel('Angle of Attack (deg)');
ylabel('Drag Coefficient (C_d)');
title('C_d vs Angle of Attack');
legend('Location', 'northeast');
grid on;

```

```

% Cm_le_alpha
Cm_LEth = [-0.13630, -0.22906, -0.32085, -0.41066, -0.49752, -0.58047];
figure;
errorbar(AoA_vals, Cm_le, dCm_le, '-go', 'LineWidth', 2, 'DisplayName', 'Experimental');
hold on;
plot(AoA_vals, Cm_LEth, '--r.', 'MarkerSize', 20, 'DisplayName', 'Theoretical');
xlabel('Angle of Attack (deg)');
ylabel('Moment Coefficient (Cm,LE)');
title('Cm,LE vs AoA');
legend('Location', 'southwest');
grid on;

```

```

% Cm_c/4_alpha
Cm_c4th = [-0.10273, -0.10438, -0.10666, -0.10954, -0.11299, -0.11697];
figure;
errorbar(AoA_vals, Cm_cby4, dCm_cby4, '-o', 'LineWidth', 2, 'DisplayName', 'Experimental');
hold on;
plot(AoA_vals, Cm_c4th, '--r.', 'MarkerSize', 20, 'DisplayName', 'Theoretical');
xlabel('Angle of Attack (deg)');
ylabel('Moment Coefficient (Cm,c/4)');
title('C_m,c/4 vs Angle of Attack');
legend('Location', 'northwest');
grid on;

```

```

% Cl vs Cd
figure;
errorbar(Cd, Cl, dCl, dCl, dCd, dCd, 'or', 'LineWidth', 2);
hold on;
plot(AoA_vals, Cl ./ Cd, '-b', 'LineWidth', 2);

```

```
scatter(NaN, panelCode.Cl, 100, 's', 'filled', 'DisplayName', 'Panel Code');
xlabel('Cd');
ylabel('Cl');
title('Drag Polar');
legend('Location', 'southeast');
grid on;
saveas(gcf, 'ee_7.png');
```

```
% L/D vs alpha
figure;
LbyD = Cl ./ Cd;
errorbar(AoA_vals, LbyD, dLbyD, '-r*', 'LineWidth', 2);
xlabel('Angle of Attack (deg)');
ylabel('Lift-to-Drag Ratio (Cl / Cd)');
title('Cl / Cd vs AoA');
legend('Experimental', 'Location', 'southeast');
grid on;
```